A Study of Fluorine Dose and Implant Energy to the NBTI upon *p*+ Implant Sequence

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***Abstract*— Negative Bias Temperature Instability (NBTI) is one of the major issues related to *p*-channel metal-oxide-semiconductor (PMOS) reliability that has been discussed for more than 40 years. In this paper, we discussed the effect of fluorine (F) co-implant to the NBTI improvement, capacitance and silicon oxide (SiO_2) thickness in term of F concentration, implant energy and also implant sequence at *p*+-region. Results suggest that, besides F dose and implant energy that is known to contribute to NBTI behaviors; implant sequence also plays a role in NBTI degradation.**

Keywords—NBTI; fluorine; implant sequence; dose; energy

I. INTRODUCTION

NBTI has been observed as early as 1967[1]. It occurs as the negative gate bias is applied on PMOS at elevated temperature. The physical mechanisms behind this NBTI degradation have been studied thoroughly by previous researcher. It is believe that the generation of traps at the Si-SiO_2 interface during negative bias temperature has further increase the NBTI degradation. Not only that, gate material, gate length, oxide electrical field and many more variables also have been studied by many researchers [2].

On top of that, the role of impurity to the NBTI degradation also has been studied in details, such as hydrogen, deuterium, nitrogen, nitrides, F and so on [3-5]. Some impurities are known to enhance the NBTI and some are help to reduce the NBTI. F is one of the impurities known to reduce the NBTI [6]. Therefore, in this study F is implanted prior to boron fluoride (BF_2) in *p*-channel of PMOS to suppress the NBTI effect.

II. EXPERIMENT

In this experiment, *p*-channel was implanted with two continuous implant species, which are F and BF_2. To study the effect of implant sequence to the NBTI degradation, one sample is implanted with F prior to BF_2 implant (standard sequence) and another sample is implanted with BF_2 followed by F (reverse sequence). Since F is known to suppress the NBTI, F dose has been diversified for both sequences and one sample is implanted without F co-implant as shown in Table 1.

TABLE I. FLUORINE DOSE IMPLANT VARIATIONS

Group	Process	Conditions		
		Dopant	Energy (keV)	Dose (cm^{-2})
Standard Sequence (F-BF_2)	Fluorine Implant	F	30	3.00E+15
	P+ Implant	BF_2	40	5.00E+15
Standard Sequence (F(10%)-BF_2)	Fluorine Implant	F	30	3.00E+14
	P+ Implant	BF_2	40	5.00E+15
Reverse Sequence BF_2-F)	P+ Implant	BF_2	40	5.00E+15
	Fluorine Implant	F	30	3.00E+15
Reverse Sequence (BF_2-F(10%))	P+ Implant	BF_2	40	5.00E+15
	Fluorine Implant	F	30	3.00E+14
BF_2	P+ Implant	BF_2	40	5.00E+15

To study the impact of F implant energy as well as the implant sequence to the NBTI degradation, F implant energy has been reduced from 30keV to 15keVfor both sequences. NBTI test was done at temperature 175°C with -5.5 V gate bias for 30 minutes. The stress was interrupted for every 200 sec for threshold voltage measurement. Measurement of threshold voltage (V_T) is done by sweeping the gate from -3V to 0V, drain bias at -3.3Vand source is grounded.

III. RESULTS AND DISCUSSIONS

Fig.1 shows the threshold voltage (V_T) shifts for different F dose and implant sequence. Standard sequence with F 100% dose (F-BF_2) is revealed to have the lowest voltage shifts and followed by reverse sequence with F 100% dose (BF_2-F). Subsequently, followed by F 10% dose for both sequences and the worst is for sample without additional F implant.

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Secondary Ion Mass Spectroscopy (SIMS) results from [7] show that F-BF_2 have slightly higher F concentration than BF_2-F that explained on the lower NBTI of F-BF_2. As we increase the F dose, there will be more Si-F bonding created thus reduce the V_T shift [6].

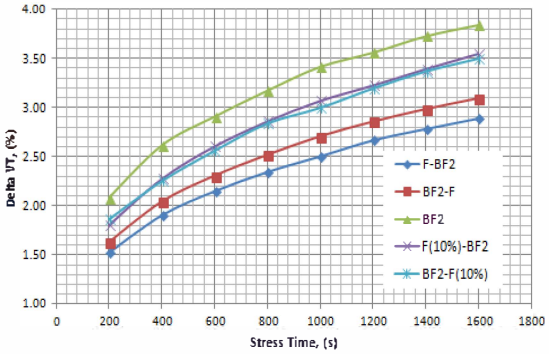

Fig. 1: Delta V_T of PMOS over the stress time for standard and reverse sequence at different F dose.

Despite a better NBTI degradation seen in higher F concentration samples, the introduction of F to the SiO_2 will increase the thickness of SiO_2 as shown in Fig. 2 due to the oxidation of oxygen released from the interaction of F with Si-O bonds, thus reduce the capacitance [8].

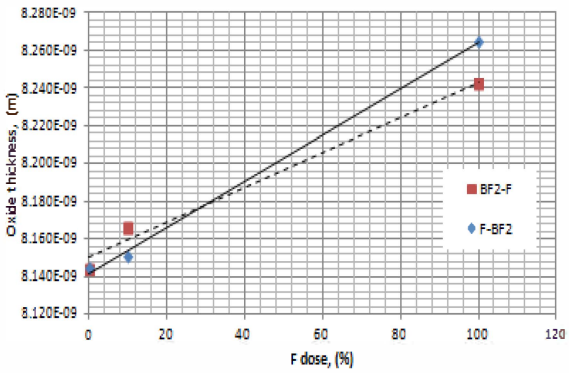

Fig. 2: SiO_2 thickness for standard and reverse sequence at different F dose.

Fig. 3 shows the NBTI degradation effect for different F implant energy.

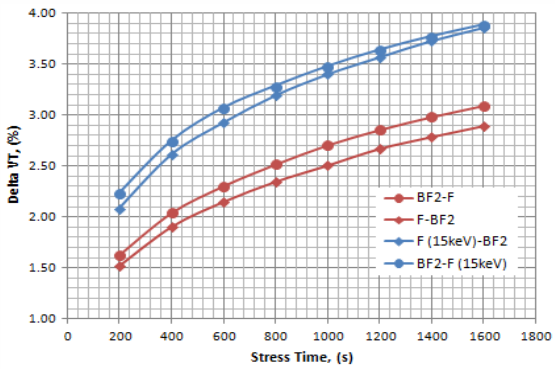

Fig. 3: Delta V_T of PMOS over the stress time for standard and reverse sequence at different F implant energies.

Consistently with observation seen in F dose variation, standard sequence shows a lower delta of V_T than reverse sequence for the same F implant energy. NBTI degradations get worst as we reduce the F implant energy. As F implant energy reduces, shallower F projected range (R_p) will be created. Therefore less F will get into the SiO_2 interface. As a result, higher delta of V_T is seen in F 15keV. At higher F implant energy, more F will gets into the SiO_2, thus increase the oxide thickness as shown in Fig. 4.

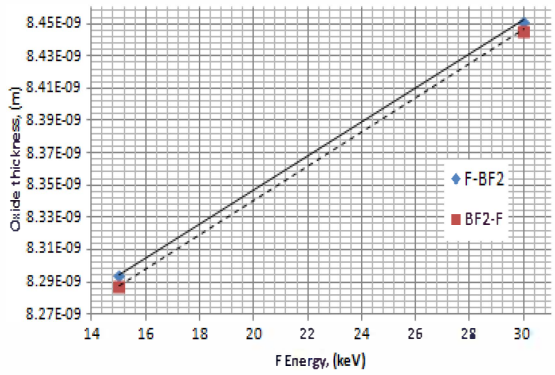

Fig. 4: SiO_2 thickness for both sequences at different F implant energies.

IV. Conclusion

From the results seen in this study, besides F concentration and implant energy, implant sequence also plays a role in the NBTI degradation. F implant prior to BF_2 implant will further increase the NBTI degradation. However, the introduction of F will increase SiO_2 thickness and reduces the capacitance.

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